Unit 3: GIS Data Model

3.1 Raster Model and Structure

A Raster Model represents geographic data as a matrix of cells (pixels), each with a value representing a specific attribute like temperature, elevation, or land cover.

- Each cell has a uniform size and stores one value.
- Ideal for continuous data such as satellite images, elevation, and rainfall.
- Simple data structure, suitable for overlay and spatial analysis.

Example: A satellite image showing land cover types where each pixel represents a land category.

Features:

- **Cell-based structure:** Data is organized in rows and columns.
- **Uniform size:** Each cell covers the same ground area.
- **Single value per cell:** Each cell stores one attribute value.

3.2 Vector Representation

The Vector Model represents geographic features using geometric shapes such as points, lines, and polygons. It is ideal for mapping discrete features with precise boundaries and locations.

Types of Vector Representation:

- **Points:** Represent single locations (e.g., trees, wells, schools)
- **Lines:** Represent linear features (e.g., roads, rivers, pipelines)
- **Polygons:** Represent area features (e.g., lakes, parks, land parcels)

Advantages of Vector Model:

- **High precision:** Accurately captures location and shape of features
- **Efficient attribute storage:** Each feature can store detailed data in an attribute table
- **Supports topology:** Maintains spatial relationships (e.g., adjacency, connectivity)

Example:

A city map showing roads as lines, parks as polygons, and schools as points.

3.3 Surface Representation in Raster Model

The raster model is commonly used to represent continuous surfaces such as terrain, temperature, or rainfall. It divides the surface into a grid of uniform cells, where each cell holds a specific value.

Surface Representation Techniques:

- **Digital Elevation Models (DEM):**
  Represent terrain surfaces by storing elevation values in each cell.
- **Digital Terrain Models (DTM):**
  Enhanced DEMs that may include features like break lines and ridges for more accurate terrain representation.
- **Satellite Imagery and Remote Sensing Data:**
  Capture surface attributes such as vegetation cover, land temperature, or soil moisture in raster format.
- **Raster Interpolation:**
  Estimates surface values at unsampled locations using methods like Inverse Distance Weighting (IDW), Kriging, or Spline.

Applications:

- **Environmental and climate studies:** Raster is ideal for modeling phenomena like rainfall, soil moisture, or temperature gradients.
- **Digital Elevation Models (DEMs):** Represent terrain by storing elevation values for each cell.

3.4 Surface Representation in Vector Model

In the **vector model**, surfaces are represented using geometric features such as **contour lines**, **TINs (Triangulated Irregular Networks)**, and **polygons**. This approach is ideal for applications requiring high accuracy and detailed representation.

Surface Representation Techniques:

Contours:

- Lines connecting points of equal value (e.g., elevation).
- Widely used for topographic and terrain mapping.

TINs (Triangulated Irregular Networks):

- Constructed from irregularly spaced elevation points connected by triangles.
- Allow variable detail and are more efficient than grids for complex terrains.

    - Common in engineering, hydrology, and 3D surface visualization.

  Polygons:

    - Represent zones of uniform surface characteristics such as land use, vegetation type, or soil type.
    - Suitable for thematic mapping and surface classification.

Advantages:

- High precision in representing surface features.
- Efficient data storage for complex topography.
- Better for engineering and infrastructure planning where detailed surface analysis is required.